CSE110: Programming Language I (Lab)
Summer 2025
Lab Exam - 02
Duration: 30 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

Design and implement a Java program that performs the following tasks:

- First, ask the user to enter a number **n** that is greater than 0.
- Then, take n **number of user inputs** (from 0 to 9): If the user enters a **negative digit**, display: "**Please enter a digit between 0 and 9.**" and re-prompt for the same digit. And, construct **a number** using the **n inputs in** the order of how they were entered. **Then, print the constructed number.** For example, if the user inputs: **2 1 4 3 6**, the constructed number is: **21436. [Hint: Multiply each digit by the appropriate power of 10 based on its position and add them together.]**
- Now, check whether the constructed number follows the pattern: **All digits of the number must be in strictly increasing order (Can not be equal either). If the number has only one digit, it is always considered a valid pattern.** Finally, show A message stating whether it is a **Valid Pattern** or **Invalid Pattern**. If it's an invalid pattern, also calculate and **print** how many times the pattern **broke**.

[N.B: You can not use arrays, strings, or any built-in methods except MATH.pow()]

Sample Input	Sample Output
Enter the value of n: 6 **Enter digit 1: 1** **Enter digit 2: 3** **Enter digit 3: 6** **Enter digit 4: 7** **Enter digit 5: 8** **Enter digit 6: 9**	**Constructed number:** 136789 **Pattern check:** Valid Pattern (**Explanation**: The pattern follows the sequence 1<3<6<7<8<9, with digits strictly in increasing order.)
Enter the value of n: 4 **Enter digit 1: 3** **Enter digit 2: -1** **Enter digit 2: 1** **Enter digit 3: 4** **Enter digit 4: 5**	Please enter a digit between 0 & 9. **Constructed number**: 3145 **Pattern check:** Invalid Pattern **Pattern Broken:** 1 times (Explanation: The pattern follows the **sequence**: 3 > 1 < 4 < 5. So, the pattern broke only once, at the transition from 3 to 1. Therefore, the pattern is invalid, and it broke 1 time.)